

# AMANA Solutions

## HR-OPS SYSTEM AUTOMATION ROADMAP

This roadmap outlines the strategy, architecture, and detailed system features for automating core HR processes at AMANA Solutions.

### 1. Executive Project Overview & Strategy

The goal of the AMANA HR System is to transition repetitive administrative tasks into digital workflows. The IT intern team is expected to build sustainable digital solutions for HR processes that allow for self-sufficient internal management going forward.

- **Primary Objectives:** Eliminate manual data tracking, prevent compliance risks (such as missing contract expirations), and improve employee self-service capabilities.
  - **Target Architecture:** Centralized Relational Database (PostgreSQL/MySQL) with a secure web front-end and automated messaging integrations (Google Chat).
  - To maximize efficiency and meet our goals, the interns are split as requested: **Interns 1 & 2** are dedicated to the **HR Pillar**, while **Intern 3** takes full ownership of the **Operations (OPS) Pillar**.
-

## 2. Detailed Automation Scopes & Functional Specifications

Below is the detailed breakdown of the four core pillars requested for system migration, detailing what changes from manual to digital, alongside backend requirements for the engineering team.

| HR Pillar                                | Current Manual Process (As-Is)                                                                        | Pain Points                                                                                                                                                                                                                                                     | Automated System Capability (To-Be)                                                                                                                                                                                                                                                                                                                                                                                                                  | KPI Metrics                                                                                                                                                                                   |
|------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Leave Database (Database Cuti)</b> | Employees check balances via chat; HR manually updates Excel sheets; approval flows are unstructured. | <ul style="list-style-type: none"><li>• High risk of <b>missing/miscalculated balances</b>.</li><li>• Lack of visibility on remaining leave quotas during <b>contract renewals</b>.</li><li>• HR wastes hours answering repetitive balance inquiries.</li></ul> | <ul style="list-style-type: none"><li>• <b>Self-Service Balance:</b> Real-time individual dashboard displaying remaining balance, history, and accrual rates.</li><li>• <b>Digital Approval Flow:</b> Employee requests leave via app -&gt; Supervisor gets a single-click email/chat approval link -&gt; Database auto-deducts balance upon approval.</li><li>• <b>Blackout Dates:</b> System blocks submissions during critical business</li></ul> | <ul style="list-style-type: none"><li>• 0% discrepancy between calculated balances and database values.</li><li>• Elimination of manual leave inquiry chats to HR regarding balance</li></ul> |

| HR Pillar                                      | Current Manual Process (As-Is)                                                                   | Pain Points                                                                                                                                                                                                                                                                                             | Automated System Capability (To-Be)                                                                                                                                                                                                                                                                                                                                                                         | KPI Metrics                                                                                                                                                                       |
|------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                |                                                                                                  |                                                                                                                                                                                                                                                                                                         | peaks automatically.                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                   |
| <b>2. Contract Tracking (Kontrak Expiring)</b> | HR manually filters dates in an Excel row, risking missing critical timelines or legal overstay. | <ul style="list-style-type: none"> <li>• High operational risk of <b>missing critical renewal timelines</b>.</li> <li>• Last-minute evaluations cause user and employee friction.</li> <li>• Legal risks from people working past their contract dates or active contracts not being updated</li> </ul> | <ul style="list-style-type: none"> <li>• <b>Proactive Trigger Engine:</b> Automated daily scan contract end dates.</li> <li>• <b>Tiered Alerts:</b> Sends automated reminders to HR and User at T-90, T-60, and T-30 days to prepare addendums or evaluations.</li> <li>• <b>Action Framework:</b> Integrates buttons inside alerts: [Generate Renewal Addendum] or [Initiate Offboarding Flow].</li> </ul> | <ul style="list-style-type: none"> <li>• 100% automated delivery of warning triggers before contract expiration.</li> <li>• Zero missed contract expiration deadlines.</li> </ul> |
| <b>3. Attendance &amp;</b>                     | Physical check-ins or manual sign-ins; medical notes are collected loosely                       | <ul style="list-style-type: none"> <li>• <b>Lost or untracked medical notes</b> leading to audit</li> </ul>                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>• <b>Digital Medical Note Upload:</b> Sick leave requests force a</li> </ul>                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>• 100% of medical leaves are recorded</li> </ul>                                                                                           |

| HR Pillar                                   | Current Manual Process (As-Is)                      | Pain Points                                                                                                                                                                                                                                                          | Automated System Capability (To-Be)                                                                                                                                                                                                                                                                                                                                                                                               | KPI Metrics                                                                                                                                                                                                                                                                                              |
|---------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Medical Leaves (Absensi &amp; Sakit)</b> | via chat apps.                                      | <p>failures.</p> <ul style="list-style-type: none"> <li>No automatic system warnings for <b>chronic or excessive sick days</b> that impact the salary payment.</li> <li>Time-consuming manual re-compilation of data rows required for the payroll cycle.</li> </ul> | <p>mandatory attachment upload field (PDF/Image file of doctor's note).</p> <ul style="list-style-type: none"> <li><b>Anomalous Flagging:</b> Auto-flags and alerts HR if an employee exceeds the maximum consecutive sick day limits allowed without a formal check.</li> <li><b>Export Engine:</b> One-click monthly CSV report compilation mapped directly to formatting templates required for payroll processing.</li> </ul> | <p>correctly so the balance matches the database exactly.</p> <ul style="list-style-type: none"> <li><b>100% data categorization</b> (zero uncategorized disease types allowed upon submission).</li> <li>HR can view successfully charts “frequent diseases” dynamically from database logs.</li> </ul> |
| <b>4. Birthday Broadcast</b>                | HR manually checks the list at the beginning of the | <ul style="list-style-type: none"> <li>Regular <b>delayed or forgotten announcements</b></li> </ul>                                                                                                                                                                  | <ul style="list-style-type: none"> <li><b>Zero-Touch Broadcasts:</b> Cron</li> </ul>                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>100% on-time automation for</li> </ul>                                                                                                                                                                                                                            |

| HR Pillar                          | Current Manual Process (As-Is)                                                                                                                          | Pain Points                                                                                                                                                                               | Automated System Capability (To-Be)                                                                                                                                                                                                                                                                                                 | KPI Metrics                                                                                                                                                               |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>(Informasi Birthday)</b>        | month and posts announcements manually.                                                                                                                 | <p>when HR is overwhelmed.</p> <ul style="list-style-type: none"> <li>• High manual effort to maintain workplace cultural milestones consistency.</li> </ul>                              | <p>job triggers a webhook connecting to the public google chat announcement channel at 09:00 AM sharp on an employee's birthday.</p> <ul style="list-style-type: none"> <li>• <b>Dynamic Template Engine:</b> Alternates between preset greeting cards and randomized warm messages to ensure engagement feels personal.</li> </ul> | <p>every employee birthday broadcast without human intervention.</p> <ul style="list-style-type: none"> <li>• Dynamic rotation of at least 3+ text variations.</li> </ul> |
| <b>5 Talent Roster Integration</b> | HR manually approaches new hires via chat to fill the form for skills and certification, then manually types certificates into an Excel row one-by-one. | <ul style="list-style-type: none"> <li>• High risk of <b>missing or skipping high-potential candidates</b> due to loose approach tracking.</li> <li>• Risk of employees having</li> </ul> | Onboarding Trigger: The moment a new hire account activates, the system triggers a mandatory Competency Assessment & organized Certificate                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>• 100% submission rate of assessment data before the onboarding phase ends.</li> </ul>                                             |

| HR Pillar                       | Current Manual Process (As-Is)                                                                                                     | Pain Points                                                                                                                                                                                                                              | Automated System Capability (To-Be)                                                                                                                                         | KPI Metrics                                                                                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |                                                                                                                                    | <p>certifications that are never tracked or saved in the database.</p> <ul style="list-style-type: none"> <li>The AI Project Matcher system cannot pull real-time talent because the raw data sits in HR's private documents.</li> </ul> | <p>Upload flow, automatically syncing structured data via API directly to the AI Talent Roster database.</p>                                                                | <ul style="list-style-type: none"> <li>0% unmapped profiles (every new hire is indexed into the AI search engine within 24 hours).</li> <li>Fully working API webhooks verified between the intern's database and the AI Product System.</li> </ul> |
| 6. Recruitment & Career Website | <p>Job openings are only broadcasted via Instagram; applicants can only apply through a separate, standalone Google Form link.</p> | <ul style="list-style-type: none"> <li>Many potential candidates <b>miss out on job openings</b> if they don't follow the Instagram account.</li> <li>Lack of a single</li> </ul>                                                        | <ul style="list-style-type: none"> <li>A dedicated, active "Career" page on the AMANA website showcasing all active vacancies, where each job listing includes a</li> </ul> | <ul style="list-style-type: none"> <li><b>100% on-time deployment</b> of the functional career web portal on the official domain.</li> <li><b>Zero dead links</b></li> </ul>                                                                        |

| HR Pillar | Current Manual Process (As-Is) | Pain Points                                                        | Automated System Capability (To-Be)                             | KPI Metrics                                                                        |
|-----------|--------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------|
|           |                                | centralized "hub" where applicants can see all open roles at once. | direct "Apply Now" button mapped to its respective Google Form. | (every "Apply Now" button dynamically redirects to the correct active Google Form) |

| OPS Pillar                | Current Manual Process (As-Is)                                                                                                                                                     | Pain Points                                                                                                                                                                                             | Automated System Capability (To-Be)                                                                                                                                         | KPI Metrics                                                                                                                                   |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Payment Request</b> | Consultants/PMs submit a Google Form. Zapier notifies Ops. Ops manually reviews a Master Sheet and manually toggles status to "Complete". An Apps Script emails Partners, who then | <ul style="list-style-type: none"> <li>Relying on separate Google Forms, Zapier, Master Sheets, and Apps Scripts creates high break/error risks.</li> <li><b>Manual Ops Bottlenecks:</b> Ops</li> </ul> | <ul style="list-style-type: none"> <li>A unified <b>Payment Request Portal</b> inside the internal system.</li> <li>The request automatically triggers an in-app</li> </ul> | <ul style="list-style-type: none"> <li>Zero manual toggles required from Ops to advance statuses (system auto-routes based on form</li> </ul> |

| OPS Pillar | Current Manual Process (As-Is)                               | Pain Points                                                                                                                                                                                                               | Automated System Capability (To-Be)                                                                                                                                                                                                      | KPI Metrics                                                                                                                                                                                                                                                                                                                                                                                        |
|------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | approve by replying via email. Finance finally schedules it. | <p>must manually toggle sheet rows to change statuses.</p> <ul style="list-style-type: none"> <li>Partner approval happens loosely via email text replies, leaving no formal audit trail inside a core system.</li> </ul> | <p>Ops Review dashboard.</p> <ul style="list-style-type: none"> <li>Approved requests instantly route to a Partner's approval queue with a single-click "Approve/Reject" action, passing directly to Finance's schedule page.</li> </ul> | <p>actions).</p> <ul style="list-style-type: none"> <li>100% centralized audit trail for Partner approvals (no more searching through email replies), instead create a button that partner will only have to click on "approval" and it can trigger an email reply to ops, finance and requestor can be notified that the request has been approved.</li> <li>Zero reliance on external</li> </ul> |

| OPS Pillar | Current Manual Process (As-Is) | Pain Points | Automated System Capability (To-Be) | KPI Metrics                              |
|------------|--------------------------------|-------------|-------------------------------------|------------------------------------------|
|            |                                |             |                                     | Zapier connectors for core data routing. |
|            |                                |             |                                     |                                          |
|            |                                | •           | •                                   | •                                        |

---

### 3. 12-Week IT Intern Development Roadmap

To ensure successful project delivery by temporary intern resources, the timeline is strictly chunked into operational milestones, focusing on incremental value delivery.

#### MONTH 1: Development Sprint

##### Week 1: Foundations, Context & DB Setup

- **Main Objective:** Align on system architecture, map schemas, gather and sync with the AI team.

- **Intern 1 (HR - Lead Backend):**
  - Set up the central repository (GitHub) and local development environments.
  - Design the core PostgreSQL/MySQL relational database schema (User, Leaves, Attendance, Contracts).
- **Intern 2 (HR - Frontend/Coordination):**
  - Conduct intense context gathering on HR requirements (leave quotas, contract alert triggers, career board structure).
  - Coordinate with the **AI Product Engineer** to map out standard data structures for the Talent Roster competency onboarding assessment.
- **Intern 3 (OPS - Standalone Owner):**
  - Analyze the workflow to map out database schemas for Payment Requests (statuses: Incomplete, Pending Ops, Pending Partner, Approved, Scheduled) and Business Trip logs.

## **Week 2: Authentication, Profile Directories & Digital Intake Forms**

- **Main Objective:** Deliver the secure login gateway and profile creation capabilities.
- **Intern 1 (HR - Backend):**
  - Build secure authentication and login modules utilizing Google Workspace OAuth.
  - Develop core profile-management CRUD operations.
- **Intern 2 (HR - Frontend):**
  - Design the Employee Profile interface where new hires can view their dashboard, complete their competency assessments, and upload certificates.
  - Create the simple frontend for the AMANA Career website page (Job listing cards linked to Google Forms).
- **Intern 3 (OPS - Full Flow):**
  - Build authentication restrictions ensuring only authorized PMs/Consultants can request payments.
  - Develop the digital submission forms for Payment Requests (eliminating Google Forms) and Business Trips (requiring active Project ID validation).

## **Week 3: Core Features Development (Split Sisi User & Admin)**

- **Main Objective:** Build out operational logic and separate user workflows from supervisor/admin controls.

- **Intern 1 (HR - User Portal & Logic):**
  - Build the user-side Self-Service Leave Application Form and Attendance Tracker.
  - Develop backend calculations guaranteeing real-time balance deductions upon request.
- **Intern 2 (HR - Admin Portal):**
  - Build the HR Admin Dashboard to review all employee leave balances, view uploaded medical logs, track contract expiration rows, and export payroll attendance data under 5 minutes.
- **Intern 3 (OPS - Admin & Approval Flows):**
  - Build the Ops Review Dashboard and Partner Approval queue to replace the Master Sheet toggling and email loops in.
  - Create the single-click "Approve/Reject" interface for Partners and the final payment scheduling board for Finance.

#### **Week 4: Automation Loops, Webhooks & System Triggers**

- **Main Objective:** Eliminate manual monitoring via background automation and communication webhooks.
- **Intern 1 (HR - Automation Background Workers):**
  - Deploy automated daily cron jobs that scan the database for contract end dates and execute tiered warning loops (T-90, T-60, T-30 days).
- **Intern 2 (HR - Communications Webhook):**
  - Build and verify Google Chat webhook integration for birthday broadcasts with at least 3+ dynamically randomized text variations.
- **Intern 3 (OPS - System Webhooks & Alerts):**
  - Replace the manual Zapier/Apps Script layers seen in [image\\_8905c7.png](#) with direct, system-triggered automated notifications sent to Ops space groups and Partners when a payment request advances.

#### **MONTH 2 & 3: Refinement, Feedback Loops & Handover**

Once Month 1 finishes, development ends, and the system transitions into an active feedback loop and deployment cycle

## Month 2: UAT Testing, Small Group Rollout & Feedback Iteration

- **Internal HR & Operations UAT:** Interns run automated mock runs with the HR and Ops teams to verify that 100% of leaves are recorded and no data glitches occur.
- **Rollout Tips (Soft Launch):** Roll out the platform to a single Project Group (PG) or small division. Real consultants and PMs will use it for actual leave requests, payment requests, and onboarding assessments.
- **Feedback Iteration:** The 3 interns spend the remaining weeks monitoring error logs, catching bugs reported by the test group, and fixing UI/UX friction points before the company-wide release.

## Month 3: Training, User Manuals & Future Developer Documentation

- **End-User Training:** Interns hold live demo sessions guiding AMANA employees on how to navigate the portal, request business trips, and manage profiles.
- **User Manuals:** Produce clear, step-by-step documentation (with screenshots) for HR, Ops, and Partners on how to handle approvals and pull data exports.
- **Future Developer Documentation:**
  - Write clean API endpoint documentation and database schema dictionaries.
  - Map out integration hooks so future developers or the AI team can easily expand the system or pull data into the *Talent Roster Expertise* matcher engine smoothly

📅 Timeline Project HR System

---

## 4. Finalized Tech Stack Requirements

- **Frontend Framework:** Next.js (App Router preferred for optimized routing)
- **Programming Language:** TypeScript (Ensures type safety across data contracts and APIs)
- **Styling Engine:** Tailwind CSS (For rapid, clean, and responsive UI development)

- **Database (BE Data Layer):** PostgreSQL (For reliable relational mapping of corporate schemas, tokens, logs, and workflow states)